

GEORGE, South Africa

WMO: 688280

Lat: 34.0042S

Lon: 22.3842E

Elev: 191

StdP: 99.05

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.2	6.1	-3.7	2.8	20.3	-1.1	3.5	17.6	12.4	17.4	11.0	17.9	2.3	340	0.420

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.3	31.1	18.2	28.8	17.8	26.8	17.9	21.9	25.6	21.2	24.7	20.5	23.9	4.3	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.9	15.9	23.2	20.1	15.1	22.5	19.2	14.3	21.8	64.9	25.9	62.3	24.5	59.9	24.1	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.4	9.0	7.8	3.5	36.5	0.7	1.9	3.0	37.9	2.6	39.0	2.2	40.1	1.7	41.5		
(5)			1.9	23.7	1.0	0.9	1.2	24.4	0.6	24.9	0.1	25.4	-0.6	26.1	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.6	20.0	20.4	19.2	17.4	15.9	13.9	13.2	13.4	14.2	15.8	17.1	19.0	(6)	(7)
	DBStd	3.58	2.01	2.16	2.43	2.95	2.87	2.82	3.05	2.64	2.58	2.69	2.27	2.12	(7)	(8)
	HDD10.0	5	0	0	0	0	0	1	3	1	0	0	0	0	(8)	(9)
	HDD18.3	901	6	4	16	51	87	136	162	154	128	88	51	16	(9)	(10)
	CDD10.0	2423	310	290	286	223	183	119	104	108	128	181	212	279	(10)	(11)
	CDD18.3	278	58	61	44	24	12	4	4	3	5	11	13	38	(11)	(12)
	CDH23.3	1649	217	240	241	204	151	70	62	66	71	115	74	139	(12)	(13)
	CDH26.7	466	38	58	82	76	45	10	9	18	29	50	22	29	(13)	(14)
(14) Wind	WSAvg	3.3	3.4	3.2	3.0	2.9	2.9	3.3	3.3	3.4	3.4	3.6	3.6	3.6	(14)	(15)
(15) Precipitation	PrecAvg	713	64	55	73	62	45	44	46	66	51	76	70	63	(15)	(16)
	PrecMax	1221	214	143	290	291	197	162	166	426	210	230	257	243	(16)	(17)
	PrecMin	375	10	14	5	10	3	5	6	10	6	3	12	2	(17)	(18)
	PrecStd	181	39	29	53	50	44	30	34	67	39	53	53	46	(18)	(19)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.2	32.2	33.5	33.1	31.0	28.0	28.0	29.1	31.1	32.4	29.2	29.8	(19)	(20)
		MCWB	20.2	21.8	20.2	18.5	15.9	13.5	13.6	13.9	16.2	17.8	18.3	20.8	(20)	(21)
	2%	DB	27.1	27.9	28.6	28.9	27.4	25.3	25.0	24.9	24.4	26.1	25.1	26.2	(21)	(22)
		MCWB	20.4	20.9	19.5	17.9	15.0	13.2	12.9	12.9	15.1	17.1	18.2	19.6	(22)	(23)
	5%	DB	25.4	25.9	25.6	25.1	24.2	22.7	22.3	21.0	21.0	22.2	22.9	24.6	(23)	(24)
		MCWB	20.2	20.5	19.4	17.4	14.6	12.4	12.2	12.7	14.5	16.3	17.5	19.1	(24)	(25)
	10%	DB	24.1	24.2	23.7	22.3	21.4	19.9	19.4	18.3	18.9	20.1	21.2	23.2	(25)	(26)
		MCWB	19.5	19.7	18.9	16.8	14.5	12.2	11.7	12.4	13.8	15.9	16.7	18.4	(26)	(27)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.7	23.0	22.0	20.6	18.2	16.1	15.5	16.6	17.9	19.7	20.7	22.0	(27)	(28)
		MCDB	26.2	28.7	27.3	27.8	24.1	21.2	20.8	20.6	25.0	25.4	25.2	26.2	(28)	(29)
	2%	WB	21.8	21.9	20.8	19.4	17.0	14.8	14.4	15.2	16.5	18.2	19.3	20.9	(29)	(30)
		MCDB	24.7	25.8	25.1	25.2	23.0	20.1	19.8	18.8	21.3	23.0	22.8	24.3	(30)	(31)
	5%	WB	21.1	21.1	20.1	18.5	16.1	14.0	13.6	14.2	15.6	17.3	18.4	20.0	(31)	(32)
		MCDB	24.0	24.5	23.9	23.0	20.9	18.8	18.7	18.3	19.5	21.0	21.5	23.2	(32)	(33)
	10%	WB	20.3	20.4	19.5	17.7	15.4	13.2	12.9	13.4	14.7	16.4	17.5	19.3	(33)	(34)
		MCDB	23.1	23.4	22.8	21.2	19.7	17.8	17.8	17.3	18.1	19.6	20.4	22.3	(34)	(35)
(35) Mean Daily Temperature Range	5% DB	MDBR	8.2	8.3	8.8	9.8	10.5	10.7	10.4	10.3	9.7	9.0	8.5	8.2	(35)	(36)
		MCDBR	10.4	11.0	13.0	15.1	15.1	15.4	15.1	15.6	14.2	14.0	11.6	10.6	(36)	(37)
		MCWBR	5.1	5.2	5.9	6.7	6.2	6.4	6.3	7.0	7.0	6.8	5.9	5.3	(37)	(38)
	5% WB	MCDBR	8.8	9.4	10.3	11.6	11.8	11.9	11.8	12.0	12.1	11.4	10.0	9.0	(38)	(39)
		MCWBR	4.5	4.8	5.2	5.8	6.2	6.2	6.4	6.7	6.7	6.0	5.5	4.7	(39)	(40)
(40) Clear-Sky Solar Irradiance	taub		0.366	0.369	0.355	0.331	0.316	0.303	0.306	0.333	0.347	0.368	0.355	0.363	(40)	(41)
	taud		2.502	2.497	2.523	2.576	2.594	2.601	2.588	2.502	2.446	2.422	2.496	2.497	(41)	(42)
	Ebn at Noon		971	947	923	893	855	841	857	875	916	937	975	978	(42)	(43)
	Edh at Noon		114	110	100	85	73	68	73	89	106	118	114	115	(43)	(44)
(44) All-Sky Solar Radiation	RadAvg		7.55	6.76	5.52	4.11	3.08	2.64	2.90	3.72	5.05	6.17	7.26	7.80	(44)	(45)
	RadStd		0.26	0.21	0.26	0.18	0.16	0.15	0.17	0.16	0.24	0.31	0.32	0.42	(45)	(46)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	+1.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		N/A	N/A	+1.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page